Comparison of DQN Architectures on CartPole-v1

CleanRL Models

1 Architectures

All models use the same training setup: 500k steps, Adam (lr 2.5×10^{-4}), buffer 10k, $\gamma = 0.99$, batch 128.

Table 1: Network architectures (CartPole: obs dim 4, 2 actions).

Model	Architecture	Activations
dqn_original	obs $\rightarrow$ 120 $\rightarrow$ 84 $\rightarrow$ actions	ReLU after each hidden layer
dqn_3layers	obs $\rightarrow$ 120 $\rightarrow$ actions	ReLU after hidden layer
dqn_linear	obs $\rightarrow$ 120 $\rightarrow$ actions	none (1 hidden, all linear)
dqn_4layers_linear	obs $\rightarrow$ 120 $\rightarrow$ 84 $\rightarrow$ actions	none (all linear)

2 Results

Training dashboards (episode return, TD loss, mean Q) for each model. Same hyperparameters; only the Q-network architecture differs.

3 Summary

- **dqn_original**: Full DQN (2 hidden layers + ReLU); typically learns well and reaches high episodic return.
- **dqn_3layers**: One hidden layer + ReLU; fewer parameters, often similar performance to original.
- **dqn_linear**: One hidden layer (120), all linear; no ReLU, equivalent to one linear map but with more parameters than single-layer linear.
- **dqn_4layers_linear**: Same shape as original but no ReLU; equivalent to one linear map, usually worse than ReLU version.

(a) dqn_original (2 hidden, ReLU)

(b) dqn_3layers (1 hidden, ReLU)

(c) dqn_linear (1 hidden, linear only)

(d) dqn_4layers_linear (2 hidden, linear only)

Figure 1: DQN training dashboards on CartPole-v1 (500k steps).